Microstructure and Mechanical Properties of Novel Fe-Mn-Ni-Co Alloys for High-Temperature Applications

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Abstract

This paper presents a comprehensive study on the design and optimization of thermomechanical processing routes for non-equiatomic Fe-Mn-Ni-Co alloys intended for critical structural applications. The alloy compositions were designed using CALPHAD-based thermodynamic calculations combined with density functional theory (DFT) ab-initio modelling to predict phase stability and mechanical response under extreme loading conditions.

Experimental validation was performed through a series of compression tests at strain rates ranging from quasi-static (10 {^{3} s {^{1}) to dynamic (10^{2} s {^{1}) conditions, including cryogenic temperature testing at -196°C. The microstructure evolution was characterized using scanning electron microscopy (SEM), electron backscatter diffraction (EBSD), and transmission Kikuchi diffraction (TKD).

Results demonstrate that the optimized processing route yields a fine-grained bimodal microstructure with enhanced twinning-induced plasticity (TWIP) behavior. The alloys exhibit excellent combinations of strength (yield stress >800 MPa) and ductility (elongation >35%) at room temperature, with maintained performance at cryogenic conditions. The grain size refinement achieved through controlled groove rolling significantly contributes to the weakening of basal texture compared to conventional processing methods.

1. Introduction

Structural materials with high impact fracture toughness are of significant interest for extreme-environment applications at low temperatures, such as transportation and storage of liquefied gases, cryogenic technologies, and structural components in aerospace vehicles. The development of novel alloy systems that combine high strength with exceptional ductility remains a fundamental challenge in materials science.

High-entropy alloys (HEAs) represent a promising class of metallic materials that offer new possibilities for applications under extreme loading conditions. Unlike conventional alloys based on one principal element, HEAs consist of multiple principal elements in near-equiatomic or non-equiatomic proportions, leading to unique combinations of mechanical properties through various strengthening mechanisms including solid solution hardening, precipitation strengthening, and deformation twinning.

Recent advances in computational materials science, particularly CALPHAD (Calculation of Phase Diagrams) methods and ab-initio calculations based on density functional theory, have enabled more efficient alloy design by predicting phase stability, transformation temperatures, and mechanical response prior to experimental validation. This integrated computational-experimental approach significantly reduces development time and cost while enabling exploration of wider compositional spaces.